

Inversa: A Contamination-Resistant, Self-Scaling Benchmark of Mathematical Ability via Problem Construction

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Abstract

Standard AI math benchmarks ask a model to **solve** a problem (problem $\rightarrow$ answer). Two well-known weaknesses follow: (1) fixed public test sets **leak into training**, inflating scores above genuine reasoning (*contamination*), and (2) each benchmark eventually **saturates** as models improve, forcing ever harder human-authored problems. We study the **inverse** task — given an answer, ask the model to **construct a problem** that yields it — and ask whether it is a more trustworthy basis for measurement. We define a machine-verified score, the **Inversa Generative Score (IGS)**, computed with a sympy oracle and **no LLM judge**. Across per-experiment cohorts of 7–22 models (plus a 59-model engine leaderboard) we find: (i) forward scores are measurably inflated by contamination on our own models (+3.2%, 95% CI [+1.0, +5.7]; GSM8K vs the fresh GSM-Symbolic), whereas the inverse task shows **no systematic gap** (+0.0%, [−10.5, +11.0]) and is contamination-immune by construction; (ii) IGS **strongly agrees** with a hard, non-saturated forward benchmark (AIME) — **Spearman +0.93** [+0.65, +1.0] — so it is a *validated measure of math ability*, **not** a distinct capability (the original “generation $\neq$ solving” hypothesis is refuted) — and a **discriminant** test (E13) shows this agreement is *math-specific*, not mere general capability: IGS tracks AIME far more than a hard non-math logic control (ρ +0.93 vs +0.66, Williams $p < 0.05$; partial ρ +0.88); (iii) IGS difficulty is **self-scaling**: raising construction difficulty (Möbius $\rightarrow$ polynomial $\rightarrow$ degree-4/composite root transforms) breaks the top ceiling so that **no model reaches 100%** (the best, opus-4.8, tops out at 88%). We report several hypotheses we tested and **rejected** along the way. Conclusion: problem construction is not a *separate* ability, but it is a **contamination-resistant, auto-scaling, machine-verified way to measure the same math ability that gold-standard forward benchmarks measure** — its durable advantage is structural, not a new axis.

Part 1 — 起 / Setup: why look at the inverse of solving

1.1 Two problems with “solve the problem” benchmarks

A forward benchmark gives a problem and grades the model’s answer. This is the dominant paradigm (GSM8K, MATH, AIME, ...). It has two documented failure modes:

- **Contamination.** Public test sets are posted online and ingested during pre-training. The model can then score well by *recall* rather than reasoning. This is established empirically: GSM1k (Zhang et al., Scale AI, 2024, arXiv:2405.00332) built a fresh GSM8K-parallel set and found an overfitting gap up to **13%** for some model families; a single test-set replica in pre-training drastically lowers MATH loss; rephrased-sample contamination evades n-gram detection (arXiv:2311.04850).
- **Saturation / the treadmill.** As models improve, each benchmark’s top fills with near-100% scores and can no longer rank the best models — like a 100-cm ruler that cannot tell apart people taller than 100 cm. The field responds by authoring harder benchmarks (AIME $\rightarrow$ FrontierMath), an expensive human treadmill, and each new benchmark eventually leaks too.

1.2 The idea

Reverse the direction. Instead of “*here is a problem, give the answer,*” ask “*here is the answer, construct a problem that has it.*” Two intuitions motivate this:

- A problem **constructed fresh at test time cannot be contaminated** — there is no fixed item to leak.
- Construction difficulty can be **raised programmatically** (compose transforms, add constraints) rather than by human authoring.

The empirical questions are whether construction is (a) *real* (not itself recall), (b) *measurable and consistent*, (c) a *trustworthy* and *durable* basis for ranking models. The rest of this paper tests these directly — including hypotheses that turned out to be **wrong**.

1.3 Hypotheses and verdicts (roadmap)

#	Hypothesis	Verdict	Evidence
H1	Forward scores are inflated by contamination	Supported	E12: +3.2% [+1.0,+5.7], 8/9 models
H2	The inverse task is recall-resistant	Supported	structural + E12b: +0.0% [−10.5, +11.0]

#	Hypothesis	Verdict	Evidence
H3	Inverse “generation” is a capability distinct from solving ($\theta_{\text{gen}} \perp \theta_{\text{solve}}$)	Rejected	E10: IGS $\leftrightarrow$ AIME +0.93 (it <i>tracks</i> ability)
H4	Forward benchmarks are saturated in general	Rejected / scoped	hard forward (AIME) is not saturated
H5	Forward solving here is <i>not</i> memorized (early over-claim)	Rejected	weak test; corrected by H1/E12
H6	IGS measures the same ability as a gold-standard hard forward test	Supported	E10: +0.93 [+0.65,+1.0]
H7	IGS difficulty can be scaled to discriminate the frontier	Supported	E8: ceiling broken, opus 88%
H8	IGS is <i>math-specific</i> , not merely general capability (g)	Supported	E13: $\rho(\text{IGS}, \text{AIME}) = .93 \gg \rho(\text{IGS}, \text{non-math logic}) = .66$, Williams $p < .05$; partial .88
H9	IGS’s construction residual predicts construction/verification beyond forward solving (incremental validity)	Rejected	E14: partial $\rho(\text{IGS}, Y \text{AIME}) \leq 0$ (Y1 -.20, Y2 -.42) — AIME predicts as well or better

The honest narrative is: the *strong* form of the original vision (a brand-new ability axis) is **false**, but a *more defensible* form (a contamination-proof, self-scaling, validated measure) **holds**.

Part 2 — / Method: exactly how Inversa works

This section is deliberately concrete so a first-time reader can see every step: what we send the model, what it returns, and how a machine (not a human, not another LLM) decides if it is correct.

Inversa scoring pipeline: answer -> constructed problem -> machine verification

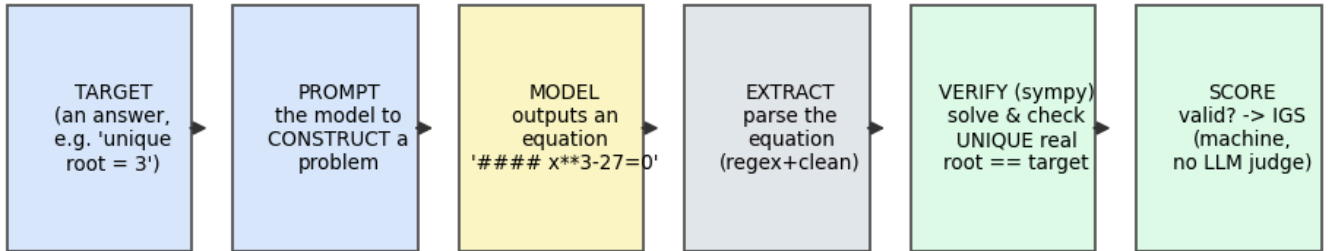


Figure 1: Inversa scoring pipeline

2.1 The two construction tasks

Task A — “Pose” (construct from a target). We give the model a target *answer* and ask it to build an equation whose **unique** real solution is exactly that target. The exact prompt is:

Construct a single-variable equation in x whose UNIQUE real solution is exactly {target}. There must be no other real solution. Radicals, fractions, exponentials and logs are allowed and encouraged.

You may reason first, but you **MUST** end with the equation on its own final line, prefixed by '#### ', in sympy syntax. Example: '#### $x**2 - 2 = 0$ '.

Targets range from easy (3) to deliberately un-memorizable ($\sqrt{2} + 1$, the real root of $t**3 - t - 1 = 0$, $\sqrt{2} + \sqrt{3}$). Why irrational targets matter: the obvious template for $\sqrt{2} + 1$ is its minimal polynomial $x**2 - 2x - 1 = 0$,

but that **also** has the conjugate root $1 - \sqrt{2} \rightarrow$ it fails “unique,” so a model must genuinely construct (e.g. $x = \sqrt{2} + 1$, or a domain-restricted form), not pattern-match.

Pose targets can be drawn from a fixed bank or generated **fresh at test time** (`--pose-random N`): the engine samples random irrational/structural targets (each target’s value is computed by sympy from the same symbolic form that describes it, so the description cannot drift from the answer), and records the seed + generated targets in the output. This makes the pose axis **contamination-immune like the transform axis** — with fresh targets there is no fixed (target $\rightarrow$ good-equation) pair to leak — closing the one asymmetry where a *fixed* pose bank could in principle be memorized once public.

A second hardening, `--pose-no-trivial`, blocks **gaming**: a linear equation $x = \text{target}$ is *correct but not construction* (it just restates the answer), so the verifier flags any degree-1-in- x answer (`trivial`) and strict mode scores it invalid — forcing degree ≥ 2 or a non-polynomial form where the target emerges as the unique real root. The effect is large (a small probe halved a frontier-mini model’s permissive pose validity once restatements were excluded), which means permissive pose is partly inflated by answer-copying; we therefore report it as an **opt-in** rigor mode (the headline runs used permissive scoring, so their pose half should be read with this caveat).

Task B — “Transform” (the recall-proof core). We generate a **random** cubic with one ugly (irrational) real root r , show it to the model, and ask it to build a *new* equation whose unique real solution is a function $g(r)$ — without computing r numerically. Exact prompt:

```
Let r be the UNIQUE real solution of this equation:
{source}                                     e.g.  x**3 + x - 1 = 0
Do NOT solve for r numerically. By transforming the structure, construct a NEW single-variable
equation in x whose unique real solution is exactly {g} (where r is that solution above).
You may reason first, but you MUST end with the new equation on its own final line prefixed by
'#### ', in sympy syntax. Example: '#### (x-1)**3 + (x-1) - 1 = 0'.
```

For $g = r + 1$ the structural answer is the substitution $x \rightarrow x - 1$: $(x-1)**3 + (x-1) - 1 = 0$. Because the source is **random** and r is irrational, **no memorized problem can supply the answer** — the output must be a function of the input shown at test time. This is what makes Task B contamination-proof by construction.

2.2 Extracting the equation from the reply

Models reply in varied formats. `extract_equation` (in `tasks/posing.py`) is robust:

1. Prefer the text after the last `####` marker.
2. Otherwise take the **last line that actually parses** as an equation.
3. Clean cosmetic noise: strip LaTeX ($\$, \backslash()$), code fences, a leading `Final: f(x) = label`; convert $\wedge \rightarrow **$ (sympy reads $\wedge$ as XOR); normalize unicode ($x^3 \rightarrow x**3$, $-\times/\sqrt{}$).
4. If the reply contains no parseable equation, **re-prompt once** for just the bare equation.

This matters: without it, a correct-but-verbose model is scored as a false zero (we observed a model drop to 0% purely from format non-compliance before this fix).

2.3 Verification — a sympy oracle, no LLM judge

`verify_equation(equation, target)` (in `verifiers/math_equation.py`) decides correctness mechanically:

1. **Parse** `lhs = rhs`. Split on a single `=`; reject inequalities and relational syntax (`Eq(...)`). Parse each side with sympy’s `parse_expr` in a **sandbox** (`__builtins__` removed) so untrusted model output cannot execute code.
2. **Require x to be present** (reject equations that do not constrain x).
3. **Solve** `sp.solve(lhs - rhs, x)` under a **10-second timeout** (a daemon thread; sympy can otherwise hang forever on adversarial input — this once stalled a run for 46 minutes).
4. Keep the **real** solutions. `valid` = the target is among them; `unique` = the real solution set is exactly `{target}` (within $1e-6$).
5. **Numeric fallback**. If symbolic solving cannot confirm the target (e.g. a transcendental embedding like $\exp(x**3 - x - 1) = 1$), evaluate the function on a grid + bisection to confirm membership and count real roots. This rescues valid constructions sympy cannot symbolically solve.

A worked example. Model output Reasoning...
`#### x**3 - 27 = 0, target 3: extract  $\rightarrow x**3 - 27 = 0$ ; solve  $\rightarrow$  roots {3, complex, complex}; real roots {3}; valid = True, unique = True  $\rightarrow$  counts. Output  $x**2 - 9 = 0$ , target 3: real roots {3,`

-3} → unique = False → **does not count** (it also solves to -3). Everything is reproducible and inspectable; the suite has **213 passing unit tests** (verifier, scoring, extraction, and the benchmark engine) — run `python -m pytest -q`.

2.4 The score: IGS

For a model we run a fixed bank of pose targets and transform items and compute two validity rates, each in $[0, 1]$:

- **pose validity** = fraction of pose items meeting the *unique-solution* constraint;
- **transform validity** = fraction of transform items whose unique real root equals $g(r)$.

IGS = mean(pose validity, transform validity).

IGS is **generative-only**; forward solving is never part of it (it is reported separately as a saturation reference). The benchmark engine is `python -m inversa.cli_bench --models ...`, which outputs a ranked IGS leaderboard (JSON + HTML) with full per-item evidence.

Interval scaling (optional). Raw IGS is an *ordinal* index — a 0.9-vs-0.8 gap is not metrically equal across the range, and pose/transform are pooled with arbitrary equal weight. `scripts/build_irt.py` recovers the model×item correctness matrix from the run cache and fits a **Rasch (1PL)** model, placing each model on an interval **logit θ scale** ($P(\text{correct}) = \sigma(\theta - b)$) with a difficulty b and a point-biserial discrimination per item. θ is a *monotone* rescaling of IGS (Spearman θ -vs-IGS = +**1.00** on the N=46 complete-case run), so the **ranking is unchanged** but the spacing becomes meaningful — the saturated 0.98→1.00 top compresses into a large θ gap near the ceiling — and pooling the two item types into one matrix drops the 50/50 weight. Output: `data/results/igs_irt.json`. (We report IGS as the headline for readability and θ as the metrically-honest companion.)

2.5 Models, data, infrastructure

- **Models.** Per-experiment cohorts of 7–22 **models**, weak (llama-3.1-8b, command-r) through frontier (opus-4.8, qwen3.7-plus, deepseek), via OpenRouter; the engine **leaderboard (§3.5)** covers **59 models / 7 families**. Per-experiment N varies as rate limits and credit drop individual models.
- **Banks** (all sympy-verified unique): pose targets (a fixed bank, or fresh via `--pose-random`, §2.1); transform items (random ugly-root cubics × {affine, reciprocal, Möbius} maps), plus the harder polynomial/brutal transforms of §3.4. Forward comparisons use **GSM8K**, **GSM-Symbolic** (HuggingFace) and **AIME 2024+2025**; the discriminant/predictive controls (§3.6, §3.8) add a logic bank and construction/verification banks.
- **Engine.** `cli_bench` schedules every (model, item) call in one bounded pool with an **item-level resume cache** (a re-run skips calls already made) and a **fast-abort** on account-wide failures — so a crash, network drop, or credit exhaustion never discards completed work and the run is resumable.

Part 3 — 轉 / Findings (and where the story turned)

3.1 Forward scores ARE inflated by contamination (H1) — and our first over-claim (H5)

We first ran a weak self-made perturbation test (number-swaps on self-authored algebra) and *wrongly concluded* forward solving was “genuine, not memorized.” That was a mistake: self-authored items **cannot** be contaminated, so finding no gap was uninformative. We corrected it with the proper test:

For 80 templates we compared accuracy on the **GSM8K original** (in training corpora = contaminated) vs the matched **GSM-Symbolic** instantiation (same difficulty, fresh numbers, canary-marked). The gap is forward score inflation.

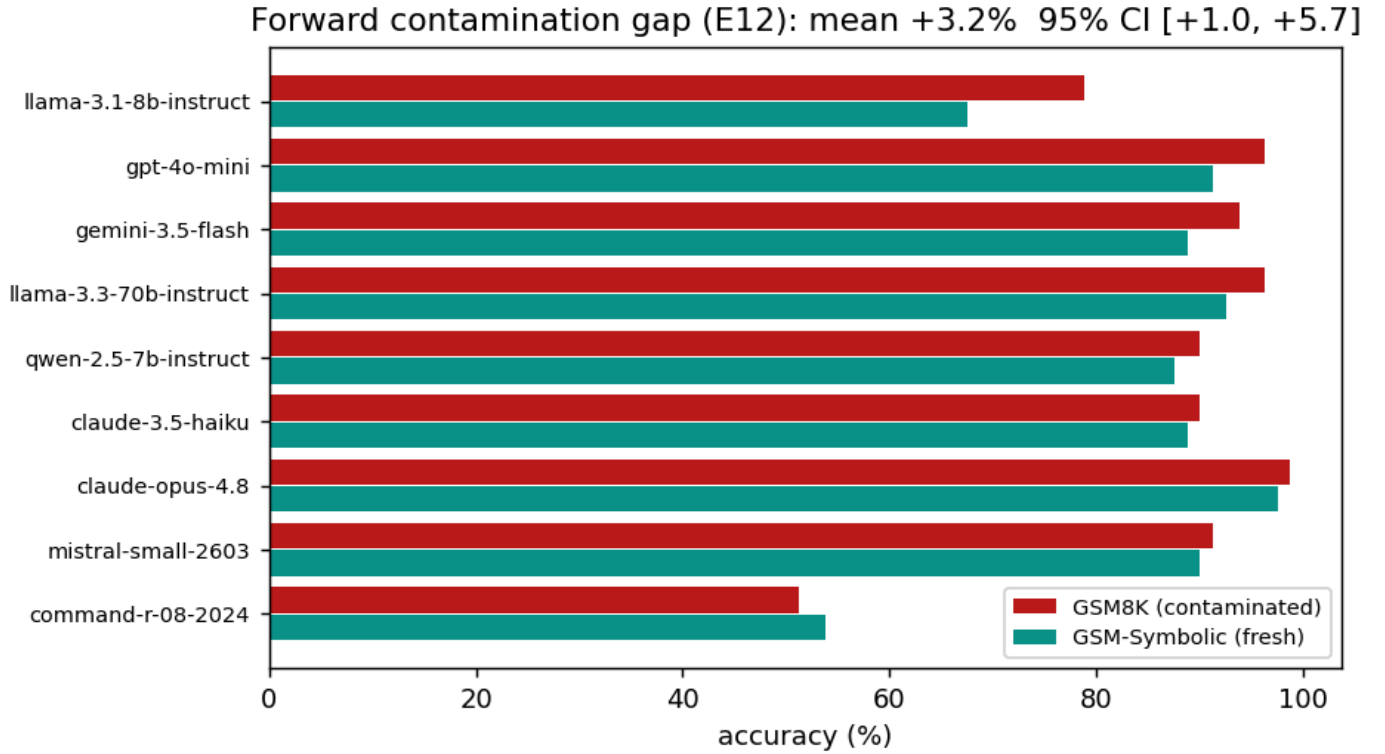


Figure 2: Forward contamination gap

Result (N=9): mean gap +3.2%, 95% CI [+1.0%, +5.7%] (excludes 0); 8 of 9 models positive. Largest for the weakest model (llama-3.1-8b +11%), small for the frontier (opus, mistral-small +1%) — exactly the model-dependent pattern GSM1k reports. So **H1 holds on our own models**: seeing the exact items inflates forward scores. (We *scope* this honestly: the effect is real but modest and uneven; we do **not** claim forward scores are broadly “fake.”)

Direct evidence (data/results/forward_gap_results.json; gap = GSM8K – GSM-Symbolic accuracy):

model	GSM8K	GSM-Symbolic	gap
llama-3.1-8b-instruct	78.8%	67.5%	+11.2%
gpt-4o-mini	96.2%	91.2%	+5.0%
gemini-3.5-flash	93.8%	88.8%	+5.0%
llama-3.3-70b-instruct	96.2%	92.5%	+3.7%
qwen-2.5-7b-instruct	90.0%	87.5%	+2.5%
claude-3.5-haiku	90.0%	88.8%	+1.3%
claude-opus-4.8	98.8%	97.5%	+1.3%
mistral-small-2603	91.2%	90.0%	+1.2%
command-r-08-2024	51.2%	53.8%	−2.5%

Independently recomputed from the file: mean +3.19%, 8/9 positive, 95% bootstrap CI [+1.0, +5.7] (20k resamples, seed 0) — matching the stored summary. See Appendix V.

3.2 The inverse task has no contamination gap (H2)

By construction, Task B uses randomized inputs, so no fixed item can leak. Empirically (E12b), we compared transform validity on **familiar/textbook** source equations ($\sqrt[3]{2}$, the plastic number, Newton’s x^3-2x-5) vs **random** sources: **mean gap +0.0%, 95% CI [−10.5%, +11.0%]** (N=7) — no systematic familiarity advantage. The CI is wide (small sample), so the **structural** argument is

primary and the measurement is corroborating. The head-to-head is the asymmetry: **forward +3.2% (CI excludes 0) vs inverse +0.0% (CI includes 0)**.

Direct evidence and an honest caveat (data/results/inverse_gap_results.json; gap = familiar – random source equations):

model	familiar	random	gap
claude-3.5-haiku	0.37	0.10	+26.7%
llama-3.3-70b-instruct	0.77	0.67	+10.0%
claude-opus-4.8	1.00	1.00	0.0%
gpt-4o-mini	0.60	0.60	0.0%
gemini-3.5-flash	0.90	0.97	–6.7%
qwen-2.5-7b-instruct	0.30	0.37	–6.7%
llama-3.1-8b-instruct	0.07	0.30	–23.3%

The mean (+0.0%) is a **cancellation of large opposite per-model swings** (–23% ... +27%), not tight agreement: with N=7 models × 30 items the per-model rates are noisy. Honesty demands flagging the claude-3.5-haiku outlier (+26.7%) — if it were not noise it would be a *familiarity advantage* for that one model. This is exactly why the **structural** argument (Task B pins the output to a randomized test-time input → there is no fixed item to leak) is the load-bearing claim; E12b only *corroborates* that the structure behaves as designed at the population level, and should not be read as a tight-CI null. Independently recomputed: mean **+0.00%**, CI [–10.5, +11.0] (Appendix V).

3.3 The big turn: IGS is *not* a separate ability — it agrees with a hard forward test (H3 rejected, H6 supported)

Our original vision claimed problem-construction is a **distinct** capability that *dissociates* from solving. To test it we needed a hard forward benchmark that is **not** saturated, and compared the model rankings. We used **AIME 2024+2025** (integer answers → exact grading; frontier models range widely).

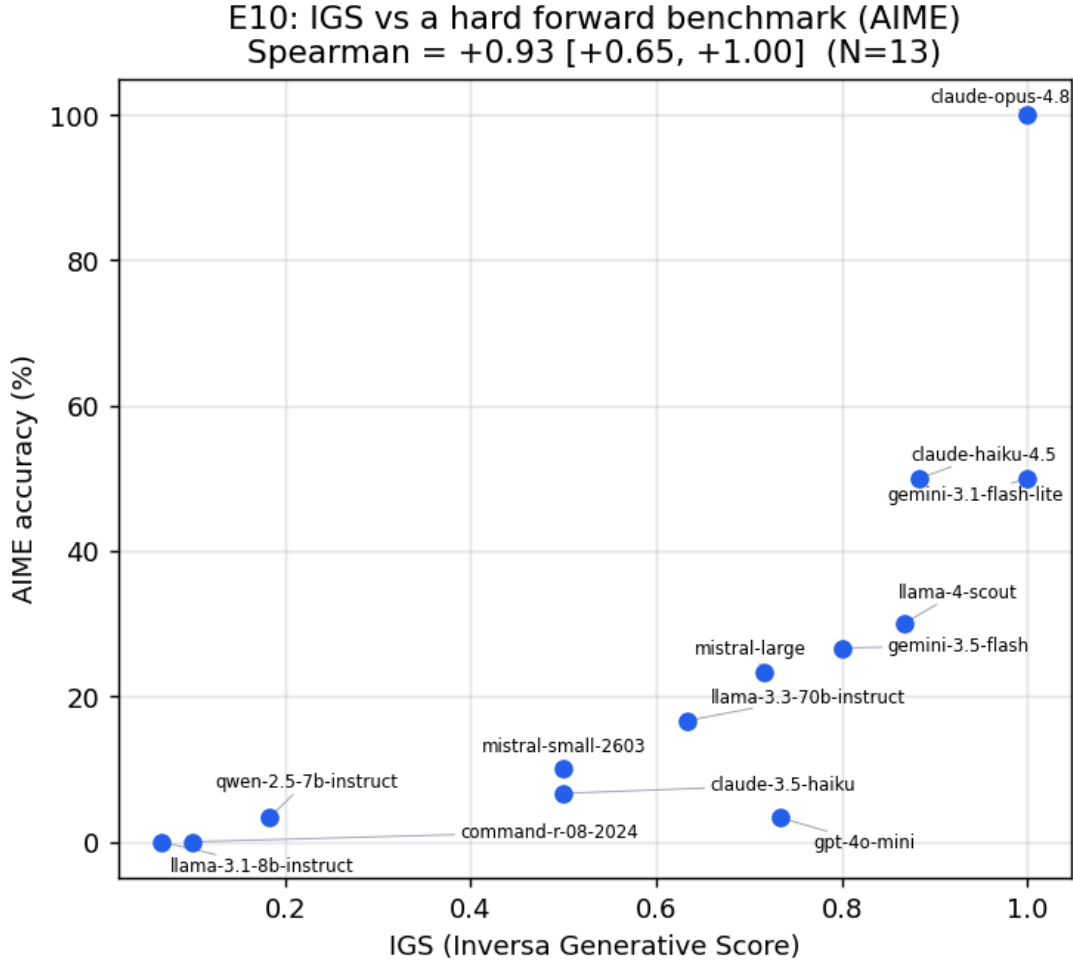


Figure 3: IGS vs AIME

Result (N=13): Spearman(IGS, AIME) = +0.93, 95% CI [+0.65, +1.0]. This is the central turn of the paper:

- The **strong** hypothesis (H3, “generation is a separate axis”) is **rejected** — a +0.93 rank correlation means IGS *tracks* general math ability, it does not measure something orthogonal.
- The **useful** hypothesis (H6) is **supported** — IGS ranks models almost identically to a gold-standard, expensive, human-authored hard benchmark. IGS is therefore a *validated measure of math ability*.

We also reject **H4** (“forward saturates in general”): AIME is *not* saturated — it discriminates models with large headroom. So Inversa’s value cannot be “forward is dead”; its value is being a **contamination-immune, auto-generated** route to the same ranking.

Direct evidence (data/results/e10_results.json; AIME = accuracy on 60 problems, AIME 2024+2025, graded by exact integer match via scripts/run_e10.py over data/banks/aime_set.json):

model	IGS	AIME
claude-opus-4.8	1.00	100%
gemini-3.1-flash-lite	1.00	50%
claude-haiku-4.5	0.88	50%
llama-4-scout	0.87	30%
gemini-3.5-flash	0.80	27%
gpt-4o-mini	0.73	3%

model	IGS	AIME
mistral-large	0.72	23%
llama-3.3-70b-instruct	0.63	17%
mistral-small-2603	0.50	10%
claude-3.5-haiku	0.50	7%
qwen-2.5-7b-instruct	0.18	3%
command-r-08-2024	0.10	0%
llama-3.1-8b-instruct	0.07	0%

Honest reading of $\rho = +0.93$. The correlation is **range-dominated**: it is anchored by the very weak models (≈ 0 on both axes) and the frontier (≈ 1 on both), so $+0.93$ *overstates* the agreement in the discriminating middle. Two facts temper the headline: (i) the **CI lower bound is $+0.65$** ($N=13$, and the upper bound is pinned at 1.0 by the percentile bootstrap) – the defensible claim is “moderate-to-strong,” not “near-identical”; (ii) there is a real **partial dissociation** – gpt-4o-mini constructs well (IGS 0.73) yet barely solves AIME (3%), ranking ~ 5 places higher on IGS than on AIME. So H3 is rejected only in its *strong* form (full orthogonality); construction and solving are *correlated but not identical*, and the residual is itself informative. Independently recomputed Spearman = **$+0.9253$** (Appendix V). The worry that this 0.93 is *just* the general-capability factor (g) is tested head-on, and rejected, in §3.6.

3.4 Difficulty is self-scaling: breaking the top ceiling (H7)

On the easy transform bank the top **saturates** – three models tie at IGS 1.0, so the very top is unmeasurable (the ruler is too short). Can we lengthen the ruler *within verifiable mathematics*? We raised construction difficulty in two steps, each verifiable via the minimal polynomial:

- **hard**: polynomial transforms ($\text{root} = r^2, r^3$) – require power-raising/elimination, not a single substitution;
- **brutal**: degree-4 / composite transforms (r^4, r^2+r, r^3-r) – deeper elimination.

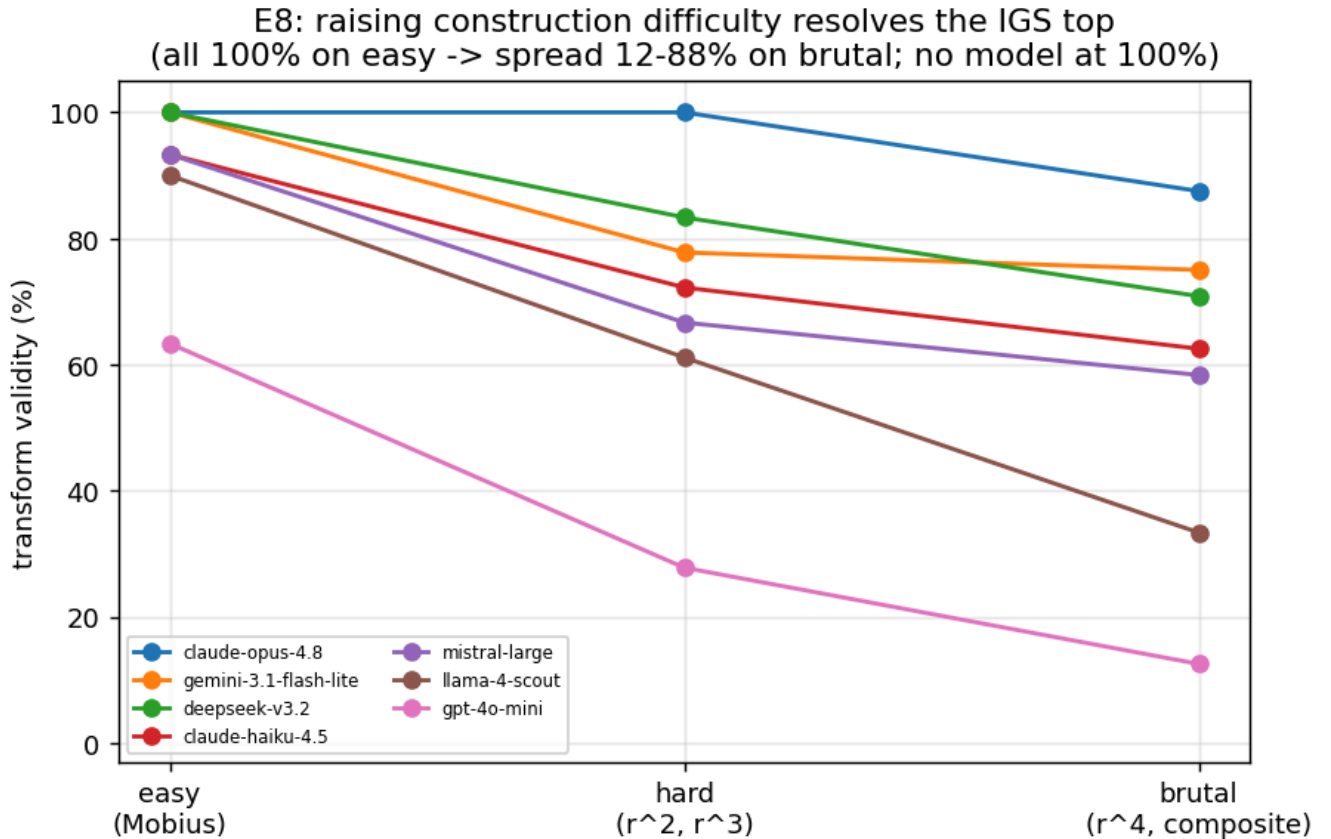


Figure 4: Difficulty escalation

Result: each step resolves more of the top. On the brutal bank **no model reaches 100%** — the best (opus-4.8) tops out at **88%**, with the field spread **12%–88%**. Recomputing IGS with the hard transform (E10-redux, N=11) **fixes the top-resolution gap** that E10 found: the number of models tied at the maximum drops from 2 to **1** (matching AIME), with **more** distinct levels (10 vs AIME’s 9) and rank-agreement maintained (**+0.89** [+0.47, +0.99]). (These redux numbers are not stored as a file but are deterministically recomputed from the committed `transform_hard_results.json` + `e10_results.json` by `python scripts/e10_redux.py` — no API — and verified to match exactly; Appendix V.) The reordering is informative, not uniform: models that do simple substitution but not elimination collapse (llama-3.3-70b 77% → 6%), showing the hard bank measures a *deeper* construction skill.

Thus the construction axis has real **headroom above saturation**, reachable programmatically and within the verifiable-math limit — the key to future-proofing (§4.3). One caveat: separating the very top two (opus vs qwen3.7-plus) was **not** achieved, because qwen3.7-plus (a slow reasoning model) times out on the hard bank; this is an infrastructure limit, not a limit of the method.

3.5 The engine-aggregated leaderboard (N=59, finer 30-item banks)

We ran the benchmark engine (`cli_bench`) over a tier-balanced roster drawn from the live OpenRouter catalogue (346 text models → 197 serious candidates → 102 attempted), scoring each on a **finer 30-item pose bank and 30-item transform bank** at **temperature 0** (deterministic scoring; a reasoning-token budget of 4000 caps spend). This finer bank supersedes the earlier coarse 6-item “standard” pose bank: pose validity now resolves in 1/30 increments rather than 1/6, so models separate on the pose axis itself rather than needing a hard-bank tiebreak.

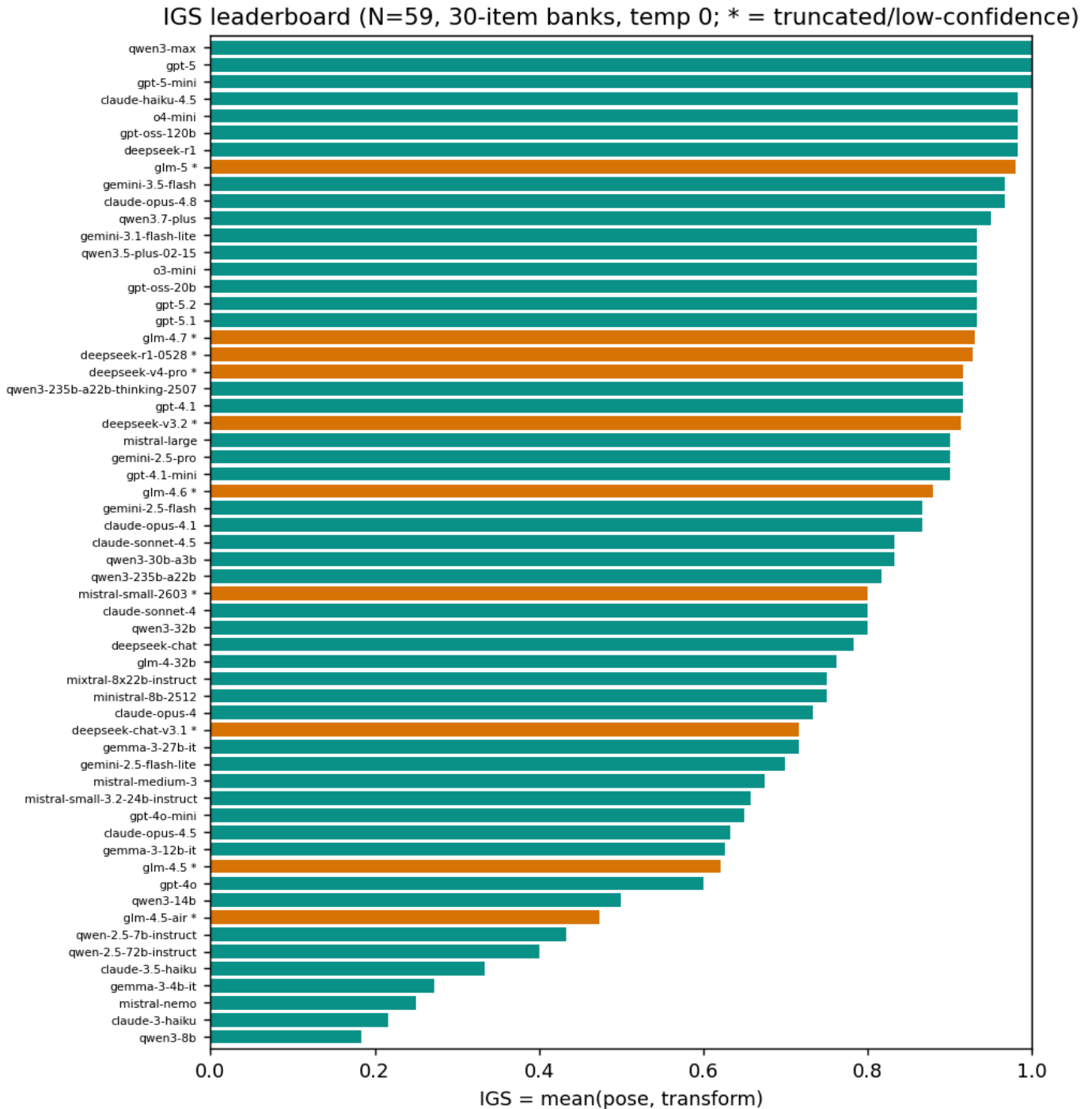


Figure 5: IGS leaderboard (30-item banks, N=59; amber * = truncated/low-confidence)

59 models across 7 families completed (IGS 0.18 → 1.00). Three models top the bank at IGS 1.0 (gpt-5-mini, gpt-5, qwen3-max) — a far tighter ceiling than the coarse bank’s five-way tie, confirming the finer bank’s higher resolution. Transform validity saturates near 1.0 for strong models while pose validity spreads 0.23 → 1.00, so the pose axis carries the discrimination. Weak open models (qwen3-8b 0.18, claude-3-haiku 0.22) anchor the bottom.

Coverage and reliability (honest). This is a single pass (one repeat, temperature 0). The account’s credit was exhausted partway through the pass; the engine detects an account-wide failure, aborts cleanly, and persists its cache, so the run is **resumable from disk** — but the remaining 43 of 102 went unmeasured: the later-roster families (meta-llama, x-ai, minimax, moonshot, cohere, amazon, nvidia, ...) were cut off, alongside the usual provider-incompatible (non-serverless / 404 / unsupported endpoint) models. Ten reasoning/large models truncated before emitting their final equation (their pose scores are low-confidence — e.g. glm-5

was scored on only 27 of 60 items); these are flagged in the dashboard. Full ranking and coverage accounting: `data/results/igs_dashboard.html` and `igs_leaderboard_30.json`. Completing the roster is a credit-bound re-run with the same `--cache` (already-measured models are skipped, so only the unmeasured set is paid for).

3.6 Discriminant validity: IGS measures *math*, not just general capability (H8)

A high IGS↔AIME correlation (§3.3) is necessary but not sufficient to call IGS a *math* measure: strong models tend to be strong at everything (a large general-capability factor, *g*), so almost any two benchmarks correlate across a wide model range. The convergent result alone cannot tell “IGS is a math test” from “IGS is a *g*-meter.” So we test **discriminant validity** — does IGS track AIME (math) *more* than a non-math axis? We ran the same E10 models on a HARD non-math control: 18 deductive **logic** ordering puzzles (no arithmetic; the prose and the answer key are generated from one constraint spec and the unique answer is **brute-force-verified**, so the key is correct by construction — `data/banks/nonmath_logic.json`). Crucially the control **spreads** the field (50%–100%), so it is a usable capability axis — unlike a factual-knowledge control, which these models saturate (we tried; even an obscure-facts bank left the field at 92%–100%, an invalid degenerate control).

Result (N=13): $\rho(\text{IGS}, \text{AIME}) = +0.93$, but $\rho(\text{IGS}, \text{logic}) = +0.66$ [$+0.17, +0.81$]; the general-capability baseline $\rho(\text{AIME}, \text{logic}) = +0.79$. IGS↔AIME *exceeds* both and survives partialling out the logic axis: partial $\rho(\text{IGS}, \text{AIME} \mid \text{logic}) = +0.88$. The 0.93-vs-0.66 gap is **statistically significant** despite the small sample (Williams’s test for two dependent correlations, $t = 3.49$, $df = 10$, $p < 0.05$). The mechanism is visible in the weak tail: **llama-3.1-8b solves the logic puzzles reasonably (78%) yet scores ≈0 on both IGS and AIME** — it can reason, but it cannot construct or solve *mathematics*, and IGS tracks the latter. So both extremes are ruled out: IGS is **not** an orthogonal new axis (H3 rejected: $\rho=0.93$ with AIME) **and not** merely general capability (H8 supported: it is specifically mathematical). Caveats: N=13 (wide CIs); a single non-math control; the top 9 models saturate the logic control, so the discriminant signal is carried by the lower-ability range; Williams’s test applies a Pearson approximation to Spearman ρ . Source/reproduce: `python scripts/run_discriminant.py --bank data/banks/nonmath_logic.json → data/results/discriminant_results.json`.

3.7 What IGS measures (the construct)

Reading the mechanism back out: IGS gives the model a *goal* (a target answer, or a transform of a random root) and asks it to **build an equation that provably and uniquely meets that goal**, which a sympy oracle then checks. To score, the model must (i) **synthesize structure backward from the goal** (generative/inverse reasoning, not forward search), (ii) **forward-solve its own construction** to know it hits the target, (iii) **reason about the whole real-solution set** to guarantee *uniqueness* (not just “a root exists”), and (iv) be **exactly** right (machine-checked — no credit for confident-but-wrong). So the construct is **goal-directed constructive formal reasoning with self-verification under an exact specification** — not calculation (the oracle does arithmetic), not recall (transform is recall-proof, `--pose-random` makes pose so too), and — per §3.6 — math-specific, not general capability.

Because step (ii) requires the ability to solve, IGS **subsumes forward solving** and shares most of its variance with it (hence E10’s $\rho=0.93$); its distinctive part is the **construction / self-verification residual** on top:

$$\text{IGS} \approx (\text{forward-solving ability}) + (\text{construction \& self-verification residual}).$$

Two caveats sharpen this. First, the construct is **mode-dependent**: permissive pose lets a model pass by *restating* the answer ($x = \text{target}$), which needs no solving at all — so permissive IGS is muddled (it is why gpt-4o-mini reaches IGS 0.73 yet only 3% on AIME); the strict mode (`--pose-no-trivial`, §2.1) removes this and isolates genuine construction. Second, construction and solving are **entangled** by the uniqueness check, so IGS cannot fully separate them — *how independently the construction residual matters* is precisely the question the predictive-validity test §3.8 puts to the data (it comes back **negative**).

3.8 Predictive validity: the construction residual does NOT predict beyond forward solving (H9 rejected)

E13 showed IGS shares math-*specific* variance with AIME; but is its distinctive construction/ self-verification residual (§3.7) *useful* — does it predict outcomes a forward-solving score does not? We tested **incremental** validity on the E10 cohort with two new machine-graded outcomes, distinct from pose/transform: **Y1** = constrained polynomial construction (build a polynomial meeting several exact constraints at once — degree, a required root, leading coefficient, distinct-real-root count — sympy-checked), and **Y2** = error detection (label yes/no whether claimed solutions/factorizations are correct). For each we ask whether IGS predicts the outcome *after controlling for AIME*.

Result (N=13): rejected — partial $\rho(\text{IGS}, Y \mid \text{AIME}) \leq 0$ for both. Y1: $\rho(\text{IGS}, Y) = +0.55$ vs $\rho(\text{AIME}, Y) = +0.66$, partial = **-0.20**. Y2: $\rho(\text{IGS}, Y) = +0.72$ vs $\rho(\text{AIME}, Y) = +0.87$, partial = **-0.42**. AIME predicts both outcomes at least as well as IGS, and once AIME is controlled IGS adds nothing — even on a *construction* outcome. So the construction residual does **not** carry demonstrated incremental predictive value over forward solving. This is consistent with the paper’s spine: IGS is not a separate ability with extra predictive power; its value is **structural** (contamination-immune, auto-scaling, math-specific ranking), not a new predictive dimension. Caveats: the cohort’s IGS here is the *permissive* easy-bank score (§3.7 argues the strict mode is a cleaner construct — untested for incremental validity); N=13 with wide CIs (Y1 CI [-0.09, +0.79]); the outcomes are themselves solving-loaded. Design + data: docs/specs/2026-06-08-e14-predictive-validity.md, data/results/predictive_results.json.

Part 4 — 結 / Conclusion

4.1 What Inversa is — and is not

- **Is:** a **machine-verified** (sympy, no LLM judge), **contamination-resistant** (forward +3.2% vs inverse ≈ 0), **self-scaling** (Möbius $\rightarrow r^2 \rightarrow r^4$ breaks the ceiling), **validated** (ranks like AIME, $\rho=0.93$) measure of mathematical ability.
- **Is not:** a *new, separate* capability (H3 rejected — it tracks general ability), nor inherently a *sharper* discriminator than a hard forward benchmark (comparable at matched difficulty).
- **Durable advantage (structural):** *verifying* a constructed problem is cheap even when *constructing* it is hard (a P-vs-NP-style asymmetry). So harder Inversa items cost a few lines of code and stay contamination-free, whereas harder forward items require expensive human authoring and eventually leak.

4.2 Threats to validity (honest)

- Moderate N (12–22) with ties $\rightarrow$ correlations carry bootstrap CIs but remain wide.
- Single domain (univariate algebra); generality untested ($\rightarrow$ §4.3).
- The contamination gap mixes leakage with surface-fragility (we did not separate them).
- The inverse-gap CI is wide; the structural claim is primary.
- Cheap verification holds for the *algebraic* regime; transcendental uniqueness can be expensive or undecidable (mitigated by timeouts + numeric fallback). On the deeper scope question this raises — whether self-scaling has a ceiling — see the *verifiability-horizon* note below; it is a boundary of evaluation itself, not a defect specific to Inversa.
- Extreme-difficulty measurement is latency-bound for slow reasoning models (qwen3.7-plus).
- Reasoning models can **truncate** (budget exhausted before the ##### eq line), which the headline runs scored as a wrong answer (a false zero that deflates the strongest models). - -truncation-missing instead treats a truncated item as **missing** (excluded from the rate, not counted wrong); the adapter flags each truncated call and the missing items are reported, so the strict mode is opt-in.

On the verifiability horizon (scope, not defect). A natural worry is that self-scaling must break where construction outruns verification — in the transcendental/undecidable regime, checking an answer can cost more than producing it. This is real, but it is a boundary of *objective evaluation itself*, and the boundary is **formal checkability, not human comprehension**. Four regimes: (a) cheaply checkable (sympy in seconds — Inversa’s sweet spot); (b) checkable but expensive (a long formal proof); (c) **beyond human reading yet machine-verified** — e.g. the Four Color Theorem and the Flyspeck proof of the Kepler conjecture are objectively certified by proof checkers no human reads end-to-end; (d) genuinely undecidable, where *no* sound terminating verifier exists. The construction>verification asymmetry *holds* through (a)–(c), so a sound checker (sympy, or a Lean/Coq proof checker) extends the scoreable frontier **far past what humans can follow** — superhuman answers can still be objectively scored as long as they are formally checkable, which is *not* an LLM judge. Only regime (d) defeats scoring — and there it defeats **every** objective benchmark, leaving only AI-judges-AI (which abandons the objectivity Inversa exists to provide; that is *peer review*, not measurement). Crucially the comparison favors Inversa: a **forward** benchmark can only pose problems **whose answer is already known**, whereas Inversa needs only to *verify* a constructed object against a spec — so its horizon (formally-checkable specs) reaches *further* than the answer-key horizon. The verifiability limit is thus where Inversa stops **later** than answer-key methods, not a weakness peculiar to it.

4.3 What remains for a stronger paper

- **E4 — generality:** replicate beyond algebra (number theory, systems, proofs) to show the construct is not algebra-specific.
- **E14 — predictive (incremental) validity: tested, rejected (§3.8).** IGS did *not* predict the construction/verification outcomes beyond AIME (partial $\rho \leq 0$). What remains: retest with the *strict* IGS (cleaner construct, §3.7), and pursue **ecological** validity

— does IGS predict a genuinely external outcome (real downstream math use, expert ratings) that no in-house math benchmark captures? That is the open form of predictive validity; the in-house incremental test has been answered.

- **E2 — reliability:** test-retest stability and inter-task consistency.
- **Larger N** to tighten all CIs; separate the very top with one more difficulty rung.

4.4 One-sentence conclusion

Problem *construction* is not a separate ability, but — because it is machine-verifiable, contamination-immune, and cheap to scale while a hard forward benchmark is not — it is a durable, trustworthy instrument that measures the same mathematical ability that gold-standard forward benchmarks measure, and keeps discriminating models as they improve.

Reproducibility

Repository `wooosshh/inversa-bench`, branch `scoring-validity`. Engine: `python -m inversa.cli_bench`. Core: `tasks/structural.py`, `verifiers/math_equation.py`, `scoring.py`. Experiments: `scripts/run_forward_gap.py` (E12), `run_inverse_gap.py` (E12b), `run_e10.py` + `e10_redux.py` (E10), `run_transform_hard.py` + `gen_transform_{hard,brutal}_bank.py` (E8), `run_discriminant.py` (E13), `run_predictive.py` (E14), `build_irt.py` (Rasch θ), `build_dashboard.py`. Figures: `scripts/make_figures.py`. Engine flags for the validity hardenings: `--pose-random` (contamination-immune pose, §2.1), `--pose-no-trivial` (anti-gaming, §2.1), `--truncation-missing` (§4.2), `--cache` (resume). All scores are sympy-verified; **213 passing unit tests** cover the verifier, scoring, extraction, and the benchmark engine (`python -m pytest -q`). Result JSONs are under `data/results/`; banks under `data/banks/`.

Appendix D — Full per-model data (embedded so the PDF verifies on its own)

These tables reproduce the actual values from `data/results/*.json` used in the text, so the paper verifies standalone. Per-model E12/E12b gaps are already in the §3.1/§3.2 body tables.

D.1 — Full IGS leaderboard (N=59, 30-item banks, temp 0; * = truncated/low-confidence)

#	model	pose	transform	IGS
1	gpt-5-mini	1.00	1.00	1.00
2	gpt-5	1.00	1.00	1.00
3	qwen3-max	1.00	1.00	1.00
4	gpt-oss-120b	1.00	0.97	0.98
5	o4-mini	0.97	1.00	0.98
6	claude-haiku-4.5	0.97	1.00	0.98
7	deepseek-r1	0.97	1.00	0.98
8	glm-5 *	0.96	1.00	0.98
9	claude-opus-4.8	0.93	1.00	0.97
10	gemini-3.5-flash	0.93	1.00	0.97
11	qwen3.7-plus	0.90	1.00	0.95
12	gpt-5.1	0.90	0.97	0.93
13	gpt-5.2	0.87	1.00	0.93
14	gpt-oss-20b	0.90	0.97	0.93
15	o3-mini	0.87	1.00	0.93
16	qwen3.5-plus-02-15	0.87	1.00	0.93
17	gemini-3.1-flash-lite	0.87	1.00	0.93
18	glm-4.7 *	0.86	1.00	0.93
19	deepseek-r1-0528 *	0.86	1.00	0.93
20	gpt-4.1	0.83	1.00	0.92
21	qwen3-235b-a22b-thinking-2507	0.83	1.00	0.92
22	deepseek-v4-pro *	0.83	1.00	0.92
23	deepseek-v3.2 *	0.83	1.00	0.91
24	gpt-4.1-mini	0.80	1.00	0.90
25	gemini-2.5-pro	0.83	0.97	0.90
26	mistral-large	0.83	0.97	0.90
27	glm-4.6 *	0.79	0.97	0.88
28	claude-opus-4.1	0.77	0.97	0.87
29	gemini-2.5-flash	0.77	0.97	0.87
30	qwen3-30b-a3b	0.70	0.97	0.83
31	claude-sonnet-4.5	0.70	0.97	0.83
32	qwen3-235b-a22b	0.67	0.97	0.82
33	qwen3-32b	0.63	0.97	0.80
34	claude-sonnet-4	0.60	1.00	0.80

#	model	pose	transform	IGS
35	mistral-small-2603 *	0.63	0.97	0.80
36	deepseek-chat	0.77	0.80	0.78
37	glm-4-32b	0.72	0.80	0.76
38	ministral-8b-2512	0.53	0.97	0.75
39	mixtral-8x22b-instruct	0.63	0.87	0.75
40	claude-opus-4	0.50	0.97	0.73
41	gemma-3-27b-it	0.50	0.93	0.72
42	deepseek-chat-v3.1 *	0.57	0.87	0.72
43	gemini-2.5-flash-lite	0.50	0.90	0.70
44	mistral-medium-3	0.48	0.87	0.67
45	mistral-small-3.2-24b-instruct	0.48	0.83	0.66
46	gpt-4o-mini	0.70	0.60	0.65
47	claude-opus-4.5	0.57	0.70	0.63
48	gemma-3-12b-it	0.55	0.70	0.63
49	glm-4.5 *	0.24	1.00	0.62
50	gpt-4o	0.57	0.63	0.60
51	qwen3-14b	0.13	0.87	0.50
52	glm-4.5-air *	0.45	0.50	0.47
53	qwen-2.5-7b-instruct	0.53	0.33	0.43
54	qwen-2.5-72b-instruct	0.53	0.27	0.40
55	claude-3.5-haiku	0.53	0.13	0.33
56	gemma-3-4b-it	0.38	0.17	0.27
57	mistral-nemo	0.40	0.10	0.25
58	claude-3-haiku	0.37	0.07	0.22
59	qwen3-8b	0.23	0.13	0.18

D.2 — Discriminant validity E13: IGS · AIME · non-math logic control (N=13)

model	IGS	AIME	logic (control)
claude-opus-4.8	1.00	1.00	1.00
gemini-3.1-flash-lite	1.00	0.50	1.00
claude-haiku-4.5	0.88	0.50	1.00
llama-4-scout	0.87	0.30	1.00
gemini-3.5-flash	0.80	0.27	1.00
gpt-4o-mini	0.73	0.03	0.94
mistral-large	0.72	0.23	1.00
llama-3.3-70b-instruct	0.63	0.17	1.00
mistral-small-2603	0.50	0.10	1.00
claude-3.5-haiku	0.50	0.07	1.00

model	IGS	AIME	logic (control)
qwen-2.5-7b-instruct	0.18	0.03	0.50
command-r-08-2024	0.10	0.00	0.56
llama-3.1-8b-instruct	0.07	0.00	0.78

D.3 — Predictive validity E14: IGS · AIME · Y1(construct) · Y2(self-verify) (N=13)

model	IGS	AIME	Y1	Y2
claude-opus-4.8	1.00	1.00	0.94	1.00
gemini-3.1-flash-lite	1.00	0.50	0.56	1.00
claude-haiku-4.5	0.88	0.50	0.75	1.00
llama-4-scout	0.87	0.30	0.75	1.00
gemini-3.5-flash	0.80	0.27	0.94	1.00
mistral-large	0.72	0.23	0.00	0.88
llama-3.3-70b-instruct	0.63	0.17	0.56	0.96
mistral-small-2603	0.50	0.10	0.31	1.00
claude-3.5-haiku	0.50	0.07	0.25	0.92
gpt-4o-mini	0.73	0.03	0.00	0.62
qwen-2.5-7b-instruct	0.18	0.03	0.44	0.79
command-r-08-2024	0.10	0.00	0.31	0.50
llama-3.1-8b-instruct	0.07	0.00	0.50	0.67

D.4 — Rasch θ (interval scaling) (N=46)

#	model	θ	IGS (raw)
1	gpt-5-mini	6.00	1.00
2	gpt-5	6.00	1.00
3	qwen3-max	6.00	1.00
4	gpt-oss-120b	4.70	0.98
5	o4-mini	4.70	0.98
6	claude-haiku-4.5	4.70	0.98
7	claude-opus-4.8	3.95	0.97
8	gemini-3.5-flash	3.95	0.97
9	qwen3.7-plus	3.49	0.95
10	gpt-5.1	3.15	0.93
11	gpt-5.2	3.15	0.93
12	gpt-oss-20b	3.15	0.93
13	o3-mini	3.15	0.93
14	qwen3.5-plus-02-15	3.15	0.93
15	gemini-3.1-flash-lite	3.15	0.93
16	gpt-4.1	2.87	0.92

#	model	θ	IGS (raw)
17	qwen3-235b-a22b-thinking-2507	2.87	0.92
18	deepseek-v4-pro	2.87	0.92
19	gpt-4.1-mini	2.63	0.90
20	gemini-2.5-pro	2.63	0.90
21	mistral-large	2.63	0.90
22	claude-opus-4.1	2.24	0.87
23	gemini-2.5-flash	2.24	0.87
24	qwen3-30b-a3b	1.92	0.83
25	claude-sonnet-4.5	1.92	0.83
26	qwen3-235b-a22b	1.78	0.82
27	qwen3-32b	1.65	0.80
28	claude-sonnet-4	1.65	0.80
29	mistral-small-2603	1.65	0.80
30	deepseek-chat	1.52	0.78
31	ministral-8b-2512	1.29	0.75
32	mixtral-8x22b-instruct	1.29	0.75
33	claude-opus-4	1.18	0.73
34	gemma-3-27b-it	1.08	0.72
35	deepseek-chat-v3.1	1.08	0.72
36	gemini-2.5-flash-lite	0.98	0.70
37	gpt-4o-mini	0.69	0.65
38	claude-opus-4.5	0.60	0.63
39	gpt-4o	0.43	0.60
40	qwen3-14b	-0.06	0.50
41	qwen-2.5-7b-instruct	-0.38	0.43
42	qwen-2.5-72b-instruct	-0.54	0.40
43	claude-3.5-haiku	-0.87	0.33
44	mistral-nemo	-1.33	0.25
45	claude-3-haiku	-1.53	0.22
46	qwen3-8b	-1.76	0.18

D.5 — Difficulty escalation E8: easy(Möbius) $\rightarrow$ hard(r^2, r^3) $\rightarrow$ brutal(r^4 , composite) (N=7)

model	easy	hard	brutal
claude-opus-4.8	1.00	1.00	0.88
gemini-3.1-flash-lite	1.00	0.78	0.75
deepseek-v3.2	1.00	0.83	0.71
claude-haiku-4.5	0.93	0.72	0.62
mistral-large	0.93	0.67	0.58

model	easy	hard	brutal
llama-4-scout	0.90	0.61	0.33
gpt-4o-mini	0.63	0.28	0.12

Appendix V — Verification & reproduction (every headline number → its source → how to recompute)

Every quantitative claim in this paper was **independently recomputed from the committed result files** (not merely read back from a stored summary), branch scoring-validity. The recomputation matches the reported values to the stated precision. The table below lets a reader verify each claim *from this document alone*: the source file, the exact stored value, the independently recomputed value, and the command to regenerate it.

Claim (§)	Source file under data/results/	Stored value	Independently recomputed	Reproduce
Forward gap +3.2% [+1.0,+5.7], N=9, 8/9 positive (§3.1)	<code>forward_gap_results.json</code> → <code>summary + models[]</code>	mean_gap 0.0319; CI [0.0097, 0.0569]; n_ positive 8	mean +3.19% , 8/9, CI [+1.0,+5.7] (20k bootstrap, seed 0)	live: <code>python scripts/run_forward_gap.py</code> ; check: <code>mean(c-f) over models[]</code>
Inverse gap +0.0% [-10.5,+11.0], N=7 (§3.2)	<code>inverse_gap_results.json</code> → <code>summary + models[]</code>	mean_gap ≈0; CI [-0.105, +0.110]	mean +0.00% ; per-model spread -23%...+27%	live: <code>python scripts/run_inverse_gap.py</code>
IGS↔AIME ρ=+0.93 [+0.65,+1.0], N=13 (§3.3)	<code>e10_results.json</code> → <code>analysis + models[]</code>	0.9253; CI [0.652, 1.0]; n 13	Spearman +0.9253 (avg-rank Pearson)	live: <code>python scripts/run_e10.py</code> (AIME from <code>data/banks/aime_set.json</code> , 60 problems)
E10-redux +0.89 [+0.47,+0.99], N=11; ties 2→1; levels 10 vs 9 (§3.4)	(<i>not stored</i>) derived from <code>transform_hard_results.json + e10_results.json</code>	—	+0.89 , tied_at_max 2→1, distinct 10 vs AIME 9	<code>python scripts/e10_redux.py</code> (no API; deterministic)
E8 brutal: opus 88% , spread 12-88% , none 100% (§3.4)	<code>transform_brutal_results.json</code> (hard_transform_validity)	opus 0.875; max 0.875; min 0.125	opus 87.5% , 12.5-87.5%	inspect per-model field (7 models)
E8 hard: llama-3.3-70b 77%→6% (§3.4)	<code>transform_hard_results.json + paper_level3_all.json</code>	easy 0.77 → hard 0.056	77% → 5.6%	inspect per-model fields
Leaderboard N=59, 7 families, IGS 0.18-1.0 (§3.5)	<code>igs_leaderboard_30.json</code> ; <code>igs_dashboard_summary.json</code>	n_models 59; igs_min 0.183	59/102, 7 families	<code>python -m inversa.cli_bench ...</code> ; <code>python scripts/build_dashboard.py</code>
Discriminant validity (E13, §3.6): $\rho(\text{IGS}, \text{AIME})=.93 \gg \rho(\text{IGS}, \text{logic})=.66$, Williams t=3.49 p<.05, partial=.88	<code>discriminant_results.json</code> (control <code>data/banks/nonmath_logic.json</code>)	strong_supported=true; williams_t 3.487; nonmath range 0.50	re-derived: gap +0.27, partial +0.88, Williams t=3.49	<code>python scripts/run_discriminant.py --bank data/banks/nonmath_logic.json</code>
Interval (Rasch) scaling (§2.4): θ monotone in IGS, θ -vs-IGS Spearman +1.00 (N=46)	<code>igs_irt.json</code>	n_models 46, n_items 60	re-derived from the run cache, no API	<code>python scripts/build_irt.py</code>
Predictive validity (E14, §3.8): partial $\rho(\text{IGS}, Y \text{AIME}) \leq 0 \rightarrow$ rejected (no incremental prediction)	<code>predictive_results.json</code>	Y1 partial -0.20, Y2 partial -0.42	re-derived: partials, gaps, spread guard	<code>python scripts/run_predictive.py</code>
Verifier soundness	<code>tests/</code>	—	213 tests pass	<code>python -m pytest -q</code>

Scope notes carried by the data (read with the numbers): (i) §3.3’s ρ is *range-dominated* — anchored by very-weak and frontier models; the defensible figure is the CI lower bound (**+0.65**; redux **+0.47**), and gpt-4o-mini (IGS 0.73 / AIME 3%) is a real partial dissociation. (ii) §3.2’s inverse-gap mean is a *cancellation* of $\pm 25\%$ per-model swings ($N=7 \times 30$ items), so the **structural** argument is primary, not the null CI. (iii) §3.4’s ceiling-breaking and §3.5’s leaderboard are partial coverage (7 brutal models with qwen3.7-plus timed out; the 59-model leaderboard is a credit-limited single pass with 10 truncated, low-confidence models flagged). All caveats are listed in §4.2.

References. GSM-Symbolic — Mirzadeh et al., Apple, ICLR 2025 (arXiv:2410.05229). GSM1k / “A Careful Examination...” — Zhang et al., Scale AI, 2024 (arXiv:2405.00332). Rephrased-sample contamination — arXiv:2311.04850.